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Test No:1 Subject:OOP's Using C++-II Date:07/01/2026 Total Marks:30

Q.1 Choose the correct alternative. (6 marks)

1. Which inheritance has one base class and many derived classes?
a) Single b) Multiple c) Hierarchical d) Hybrid
2. Pointer to base class can point to:
a) Only base object b) Only derived object c) Both base and derived objects d) None
3. A class having at least one pure virtual function is called:
a) Base class b) Derived class c) Abstract class d) Virtual class
4. Which access specifier allows members to be accessible in derived classes but not outside?
a) Public b) Private c) Protected d) Default
5. Which inheritance has many base class and one derived classes?
a) Single b) Multiple c) Hierarchical d) Hybrid
6. Which inheritance causes diamond problem?
a) Single b) Multiple c) Multilevel d) Hierarchical

Q.2 Answer the following (Any Three). (6 marks)

1. Explain benefits of inheritance.
2. Explain types of derivation: public, private, protected.
3. Explain virtual destructor and its importance.
4. Define inheritance. Explain single inheritance with example.
5. Explain hybrid inheritance.

Q.3 Answer the following (Any Two). (6 marks)

1. Explain overloaded member functions.
2. Short note on container class.
3. Write a cpp to implement Virtual destructor.

Q.4 Answer the following (Any Two). (6 marks)

1. Explain behavior of constructors and destructors in multilevel inheritance.
2. Explain virtual functions and their characteristics.
3. Write a cpp to implement multipath inheritance.

Q.5 Answer the following (Any One). (6 marks)

1. Explain abstract class and pure virtual function.
2. Explain pointer to base class and pointer to derived class.